

PHASE 3: DATA COLLECTION & DATA PREPARATION

Project Details

Particular	Details
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Phase	Data Collection & Data Preparation
Date	31 January 2025
Team ID	_____
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3.1 Data Collection & Dataset Description

Introduction

The Data Collection phase is the foundation of the analytical process. High-quality and well-structured data is essential for generating accurate visualizations and meaningful business insights. For this project, a publicly available Food Ordering Behaviour dataset was selected because it contains comprehensive information about customer ordering patterns, demographics, restaurant categories, cuisines, meal types, ratings, delivery performance, and order values.

The dataset was downloaded from Kaggle in CSV format and imported into Tableau Desktop for further processing and analysis.

Dataset Source

Attribute	Description
Dataset Name	Food Ordering Behaviour Dataset
Source	Kaggle
File Format	CSV (.csv)
Records	50,000
Columns	19

Attribute	Description
Data Type	Structured Tabular Data

Dataset Features

The dataset contains customer and order-related attributes, including:

- Order ID
- Customer Age
- Gender
- City
- Restaurant Type
- Cuisine Type
- Meal Type
- Company Type
- Order Value
- Delivery Fee
- Delivery Time
- Customer Rating
- Payment Method
- Order Date
- Order Status

These attributes enable comprehensive analysis of food ordering behaviour and customer preferences.

Objectives of Data Collection

- Obtain a reliable and publicly available dataset.
- Ensure sufficient records for meaningful analysis.
- Support multiple visualization techniques.
- Enable customer segmentation and trend analysis.

- Provide data suitable for Tableau dashboards and stories.

3.2 Dataset Loading into Tableau

Introduction

After downloading the dataset, the next step was to load it into Tableau Desktop. Tableau provides a simple interface for connecting to external data sources such as CSV, Excel, SQL databases, and cloud platforms.

The dataset was imported through Tableau's Data Source page, where all columns were verified before analysis.

Loading Procedure

1. Open Tableau Desktop.
2. Select **Connect** → **Text File**.
3. Browse and select **food_ordering_behaviour_dataset.csv**.
4. Import the dataset into Tableau.
5. Review the Data Source tab.
6. Verify column names.
7. Check data types.
8. Confirm the total number of records.

Data Verification

The following validations were performed after loading the dataset:

Validation	Status
Dataset Loaded Successfully	✓
Total Records Imported	50,000
Number of Columns	19
Duplicate Records	None Detected
Data Types Verified	Yes

Validation	Status
Missing File Errors	None

Benefits of Tableau Data Loading

- Fast connection to CSV files.
 - Automatic data type detection.
 - Easy preview of imported records.
 - Efficient handling of large datasets.
 - Seamless integration with visualization tools.
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3.3 Data Preparation & Transformation

Introduction

Raw datasets often contain inconsistencies, missing values, and formatting issues that may affect visualization quality. Therefore, the dataset was cleaned and transformed before creating dashboards.

The objective of data preparation was to ensure that the dataset was accurate, complete, and suitable for interactive analysis.

Data Cleaning Activities

The following preprocessing activities were performed:

- Verified column names.
 - Checked missing values.
 - Removed irrelevant fields (if any).
 - Ensured consistent data formats.
 - Validated numerical and categorical values.
 - Checked duplicate records.
 - Standardized categorical labels.
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Calculated Field – Age Group

A calculated field was created in Tableau to classify customers into meaningful age categories.

Customer Age Age Group

Age ≤ 20	Child
21–30	Gen Z
31–40	Millennial
Above 40	Adults

This calculated field simplified demographic analysis and enabled grouped visualizations.

Data Transformation

Several transformations were performed to improve analytical quality:

- Converted age values into age groups.
- Categorized customers by meal type.
- Standardized cuisine names.
- Organized city information.
- Verified customer ratings.
- Prepared measures and dimensions for Tableau.

Benefits of Data Preparation

- Improved data quality.
- Better dashboard performance.
- Easier customer segmentation.
- More accurate analytical results.
- Enhanced visualization clarity.

3.4 Data Validation & Exploratory Analysis

Introduction

Before developing dashboards, the prepared dataset was validated to ensure completeness and accuracy. Exploratory Data Analysis (EDA) was also conducted to understand data distribution, identify trends, and determine suitable visualization techniques.

Dataset Validation

Validation Parameter	Result
Total Records	50,000
Total Columns	19
Missing Values	Checked and handled
Duplicate Records	None
Invalid Data Types	None
Dataset Ready for Visualization Yes	

Exploratory Data Analysis

The dataset was explored to identify key trends before visualization.

Customer Analysis

- Distribution of customers across different age groups.
- Gender-wise customer participation.
- Customer ratings analysis.

Order Analysis

- Meal-wise order distribution.
- City-wise order volume.
- Company type analysis.
- Restaurant category analysis.

Revenue Analysis

- Revenue generated by meal type.
- Average order value.

- Delivery fee trends.

Delivery Analysis

- Delivery time distribution.
 - Order completion trends.
 - Customer satisfaction based on delivery performance.
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Observations

The exploratory analysis revealed several useful insights:

- Meal preferences vary across customer age groups.
- Certain cuisines are significantly more popular in specific cities.
- Dinner orders contribute a substantial portion of total revenue.
- Customer ratings generally remain high for faster deliveries.
- Company type influences meal ordering patterns.

These findings guided the selection of visualization techniques used in the next phase.

Conclusion

The Data Collection and Data Preparation phase ensured that the Food Ordering Behaviour dataset was complete, accurate, and ready for visualization. The dataset was successfully downloaded, loaded into Tableau, cleaned, validated, and transformed using calculated fields and preprocessing techniques. Exploratory analysis provided valuable insights into customer behaviour and ordering trends, establishing a reliable foundation for developing interactive dashboards and Tableau stories in the next phase.